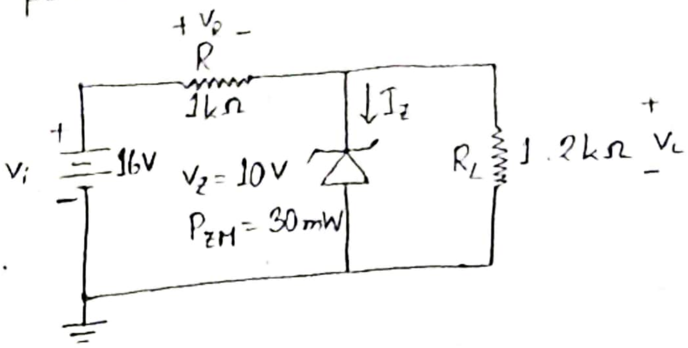
Zener Diode
Chapter : 02

Example: 2.26

(a) For the Zener diode network of figure, determine V_L, V_R, I_Z and P_Z.

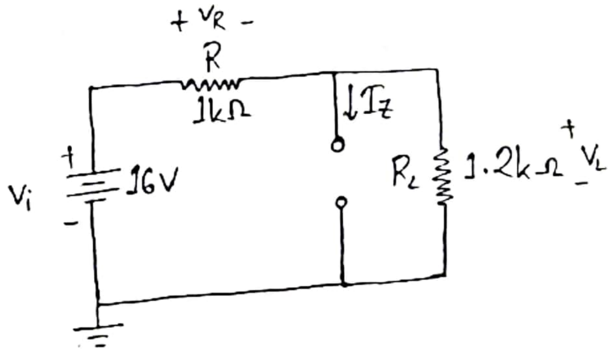
(b) Repeat part (a) with $R_L = 3k\Omega$



Solution:-

(a) $V_L = \frac{R_L}{R+R_L} \times V_i$

$= \frac{1.2}{1+1.2} \times 16 = 8.73V$

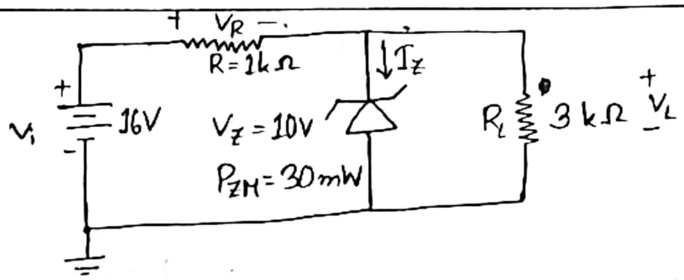


$V_R = 16 - 8.73 = 7.27V$

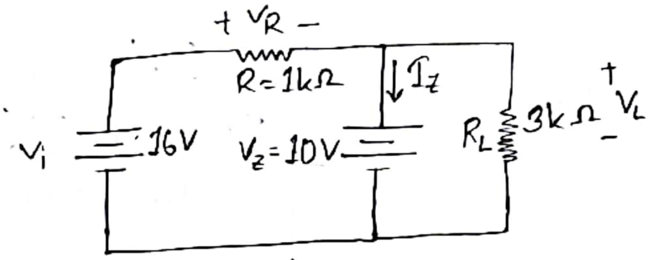
$I_Z = 0mA$

$P_Z = 0 mW$

(b)



$$V_L = \frac{R_L}{R+R_L} \times V_i$$
$$= \frac{3}{1+3} \times 16$$
$$= 12V$$



$V_L = 10V$, $V_Z = 10V$

$V_R = 16 - 10 = 6V$

$$I_R = \frac{V_R}{R} = \frac{6}{1} = 6mA$$

$$I_L = \frac{V_L}{R_L} = \frac{10}{3} = 3.33mA$$

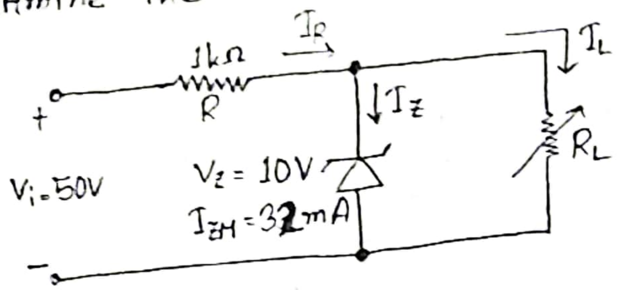
$I_Z = 6 - 3.33 = 2.67mA$

$P_Z = V_Z I_Z = 10 \times 2.67 = 26.7\ mW$

Example:
2.27

(a) For the network of figure, determine the range of R_L and I_L that will result in V_{RL} being maintained at 10V.

(b) Determine the maximum wattage rating of the diode.



Solution:-

$R = 1k\Omega$
$= 1000\ \Omega$

(a) $V_L = V_Z = \dfrac{R_L}{R_L + R} \times V_i$

$\Rightarrow 10 = \dfrac{R_L}{R_L + 1000} \times 50$

$\Rightarrow 10 = \dfrac{50R_L}{R_L + 1000}$

$\Rightarrow 10R_L + 10000 = 50R_L$

$\therefore R_{L_{min}} = 250\ \Omega$

$V_R = 50V - 10V = 40V$

$I_R = \dfrac{V_R}{R} = \dfrac{40}{1000} = 0.04A = 40mA$

$I_{max} = \dfrac{10}{250} = 0.04A$

$I_{L_{min}} = 40 - 32 = 8mA$

$R_{max} = \dfrac{10}{8} = 1.25k\Omega = 1250\ \Omega$

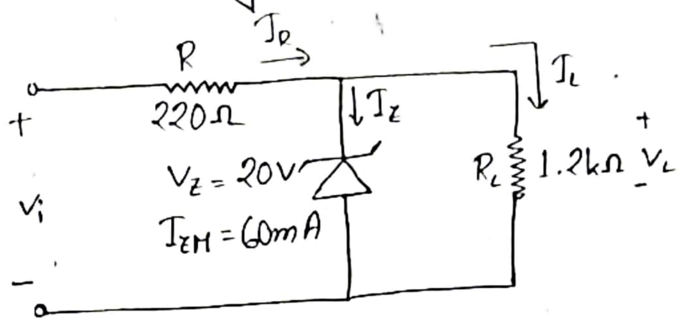
(b) $P = V_z I_{ZM}$

$= 10V \times 32mA$

$= 320mW$

Example: 2.28

Determine the range of values of V_i that will maintain the Zener diode of figure in the " on " state.



Solution:-

$R_L = 1.2k\Omega$
$= 1200\Omega$

$$V_z = \frac{R_L}{R_L + R} \times V_i$$

$$\Rightarrow 20 = \frac{1200}{1200 + 220} \times V_i$$

$$\therefore V_{i_{min}} = 23.67 \text{ V}$$

$$I_L = \frac{20}{1200} = 0.0167 A = 16.67mA$$

$$I_R = 60 + I_L = 60 + 16.67 = 76.67mA$$

$$V_R = 220 \times 76.67 = 16867.4 mV$$
$$= 16.86 V$$

$$V_{i\,max} = (16.86 + 20) V$$

$$= 36.86 V$$

BJT

Chapter : 4

Example: 4.1

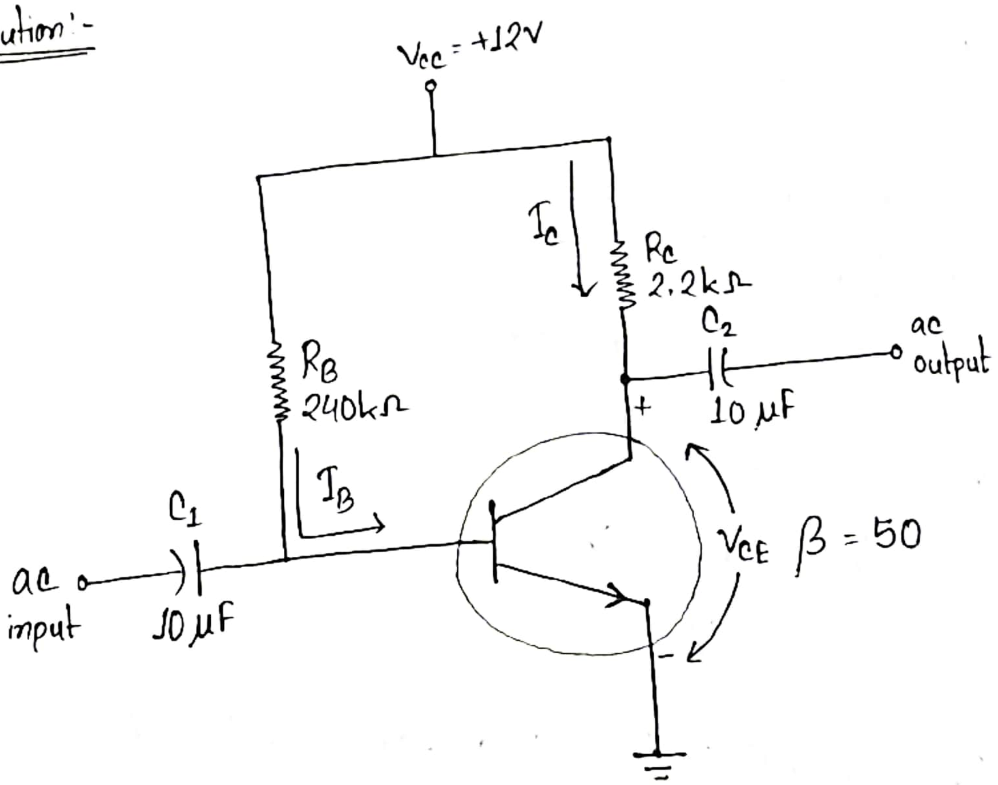
Determine the following for the fixed-bias configaration
(a) I_{B_Q} and I_{C_Q}

(b) V_{CE_Q}

(c) V_B and V_C

(d) V_{BC}

Solution:-



Applying KVL at input side,

$$-V_{CC} + I_B R_B + V_{BE} = 0$$

$$\Rightarrow I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

$$\therefore I_B = \frac{12 - 0.7}{240} = 0.047\,mA$$

$I_C = \beta I_B$

$= 50 \times 0.047 = 2.354\,mA$

Applying KVL at output side,

$-V_{CC} + I_C R_C + V_{CE} = 0$

$\Rightarrow V_{CE} = V_{CC} - I_C R_C$

$\therefore V_{CE} = 12 - (2.35 \times 2.2)$

$= 6.83\,V$

$V_{BE} = V_B - V_E$

$\Rightarrow V_B = V_{BE} + V_E$

$\therefore V_B = 0.7 + 0 = 0.7\,V$

$V_{CE} = V_C - V_E$

$\Rightarrow V_C = V_{CE} + V_E$

$\therefore V_C = 6.83 + 0 = 6.83\,V$

$V_{BC} = V_B - V_C$

$= (0.7 - 6.83)\,V$

$= -6.13\,V$

(a) $I_{BQ} = 0.047\,mA$ and $I_{CQ} = 2.35\,mA$

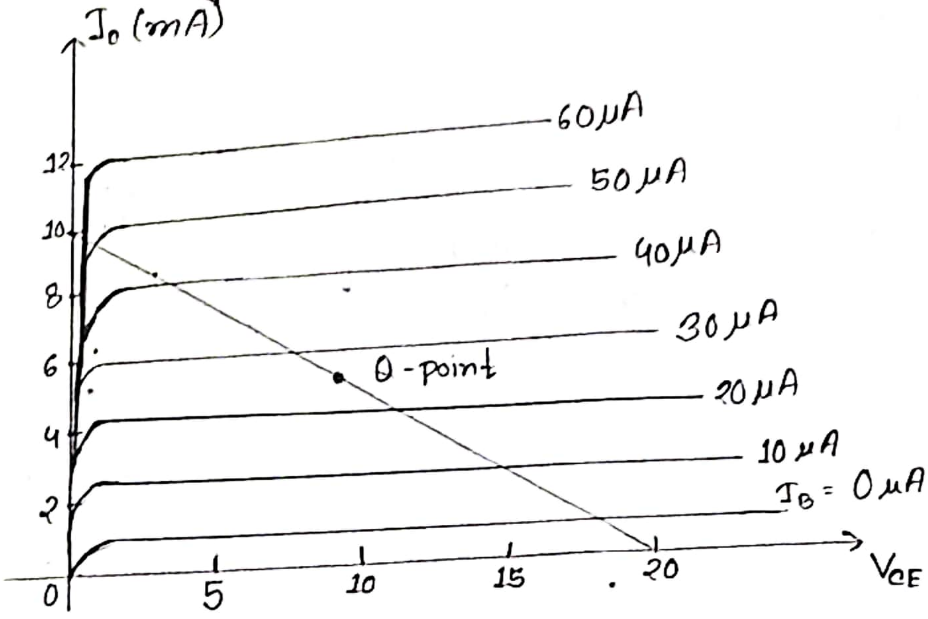
(b) $V_{CE_Q} = 6.83\,V$

(c) $V_B = 0.7\,V$ and $V_C = 6.83\,V$

(d) $V_{BC} = -6.13\,V$

Example 4.3

Given the load line of figure and the defined Q-point, determine the required values of V_{CC}, R_C and R_B for a fixed-bias configuration.



Solution:-

$$I_B = 25\ \mu A$$

$$\frac{V_{CC}}{R_C} = 10 \quad , \quad V_{CC} = 20$$

$$R_C = \frac{V_{CC}}{I_C} = \frac{20}{10} = 2k\Omega$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

$$\Rightarrow R_B = \frac{V_{CC} - V_{BE}}{I_B}$$

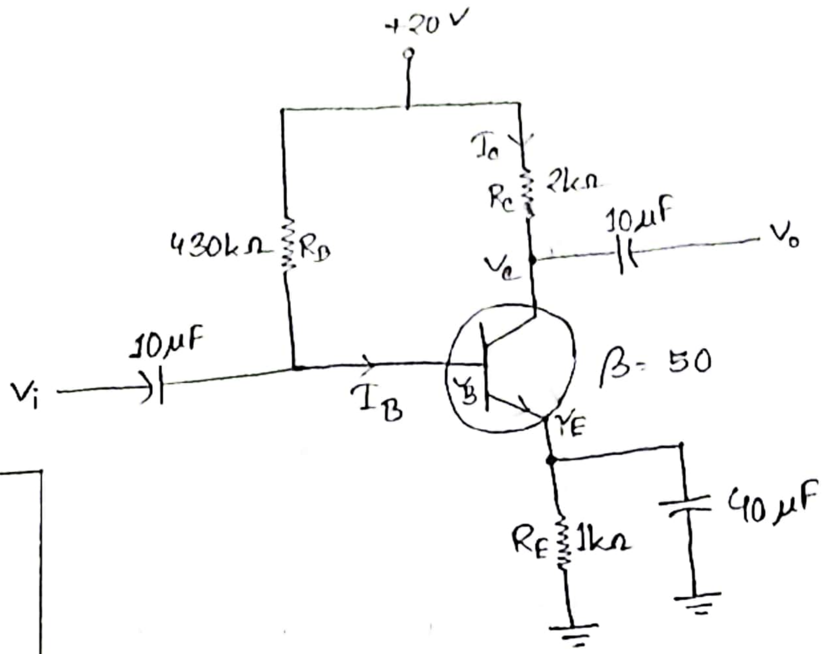
$$\Rightarrow R_B = \frac{20 - 0.7}{25\times10^{-6}}$$

$$\therefore R_B = 772\ k\Omega$$

Example: 4.4

For the emitter-bias network of figure, determine:

a. I_B
b. I_C
c. V_{CE}
d. V_o
e. V_E
f. V_B
g. V_{BC}



$$I_E = I_B + I_C$$
$$I_E = I_C$$
$$I_C = \beta I_B$$
$$I_E = I_B(\beta+1)$$

Solution:-

Applying KVL at input side,

$$-V_{CC} + I_B R_B + V_{BE} + I_E R_E = 0$$

$$\Rightarrow -V_{CC} + I_B R_B + V_{BE} + I_B(\beta+1) R_E = 0$$

$$\Rightarrow I_B \left(R_B + (\beta+1) R_E \right) = V_{CC} - V_{BE}$$

$$\Rightarrow I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta+1) R_E}$$

$$\Rightarrow I_B = \frac{20 - 0.7}{430 + (50+1) \cdot 1}$$

$$\therefore I_B = 0.04 mA$$

$I_c = \beta I_B = 50 \times 0.04 = 2mA$

Applying KVL at output side,

$-V_{CC} + I_C R_C + V_{CE} + I_E R_E = 0$

$\Rightarrow -V_{CC} + I_C R_C + V_{CE} + I_C R_E = 0$

$\Rightarrow I_C (R_C + R_E) = V_{CC} - V_{CE}$

$\Rightarrow V_{CE} = V_{CC} - I_C (R_C + R_E)$

$\Rightarrow V_{CE} = 20 - 2(2+1)$

$\therefore V_{CE} = 14\,V$

$V_C = V_{CC} - I_C R_C$

$= 20 - 2 \times 2 = 16V$

$V_{CE} = V_C - V_E$

$\Rightarrow V_E = V_C - V_{CE} = 16 - 14 = 2V$

$V_{BE} = V_B - V_E$

$\Rightarrow V_B = V_{BE} + V_E = 0.7 + 2 = 2.7V$

$V_{BC} = V_B - V_C$

$= 2.7 - 16 = -13.3V$

a. $I_B = 0.04mA$

b. $I_C = 2mA$

c. $V_{CE} = 14V$

d. $V_C = 16V$

e. $V_E = 2V$

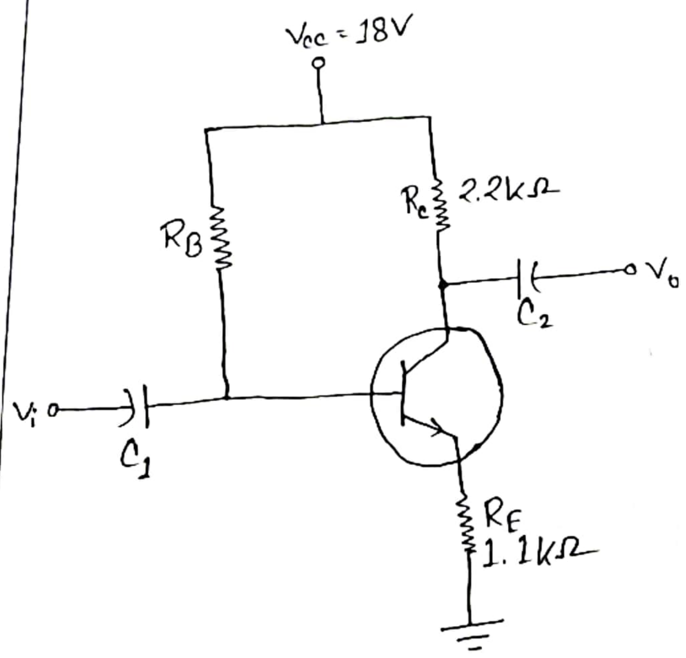
f. $V_B = 2.7V$

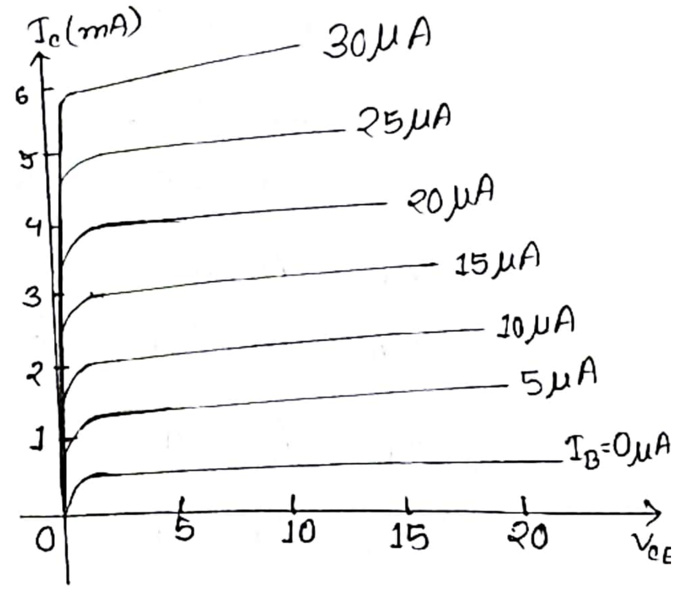
g. $V_{BC} = -13.3V$

CamScanner

Example: 4.7

a. Draw the load line for the network of figure on the characteristics for the transistor appearing in figure.
b. For a Q-point at the intersection of the load line with a base current of 15 µA, find the values of I_{CQ} and V_{CEQ}.
c. Determine the dc beta at the Q-point.
d. Using the beta for the network determined in part c, calculate the required value of R_B and suggest a possible standard value.





Solution:-

(a) Load Curve:

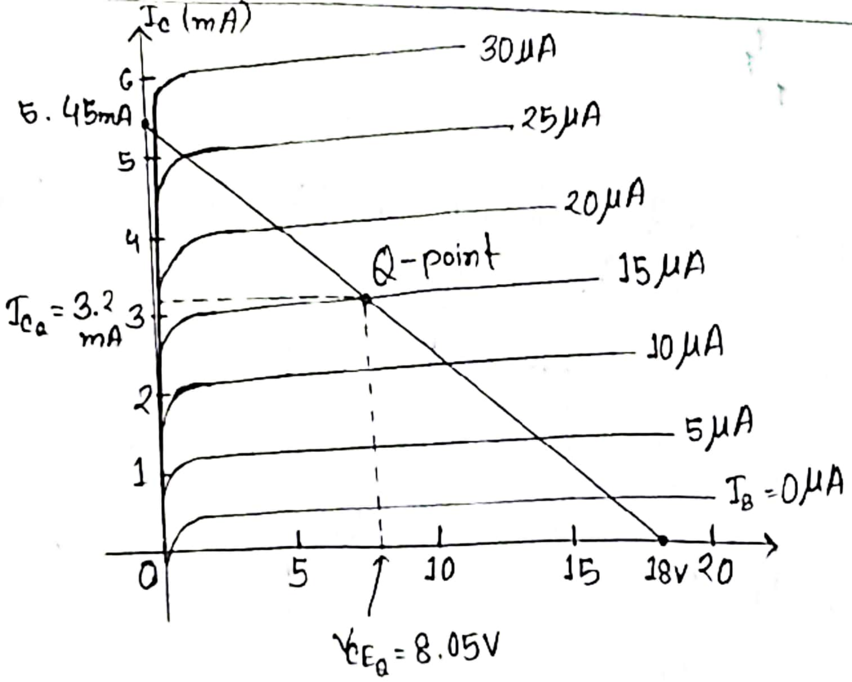
$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

When,

$$V_{CE} = 0, \quad I_C = \frac{V_{CC}}{R_C + R_E} = \frac{18}{2.2 + 1.1} = 5.45\,mA$$

When,

$$I_C = 0, \quad V_{CC} = V_{CE} = 18\,V$$



(b) From the characterics,

$$I_{CQ} = 3.2mA \; ; V_{CE_Q} = 8.05 V$$

(c) $\beta = \frac{I_{CQ}}{I_{BQ}} = \frac{3.3 \times 10^{-3}}{15 \times 10^{-6}} = 220$

d) Applying KVL at input side,

$$-V_{cc} + I_B R_B + V_{BE} + I_E R_E = 0$$

$$\Rightarrow -V_{cc} + I_B R_B + V_{BE} + I_B(\beta + 1) R_E = 0$$

$$\Rightarrow I_B = \frac{V_{cc} - V_{BE}}{R_B + (\beta+1) R_E}$$

$$\Rightarrow R_B = \frac{V_{cc} - V_{BE} - (\beta+1) R_E I_B}{I_B}$$

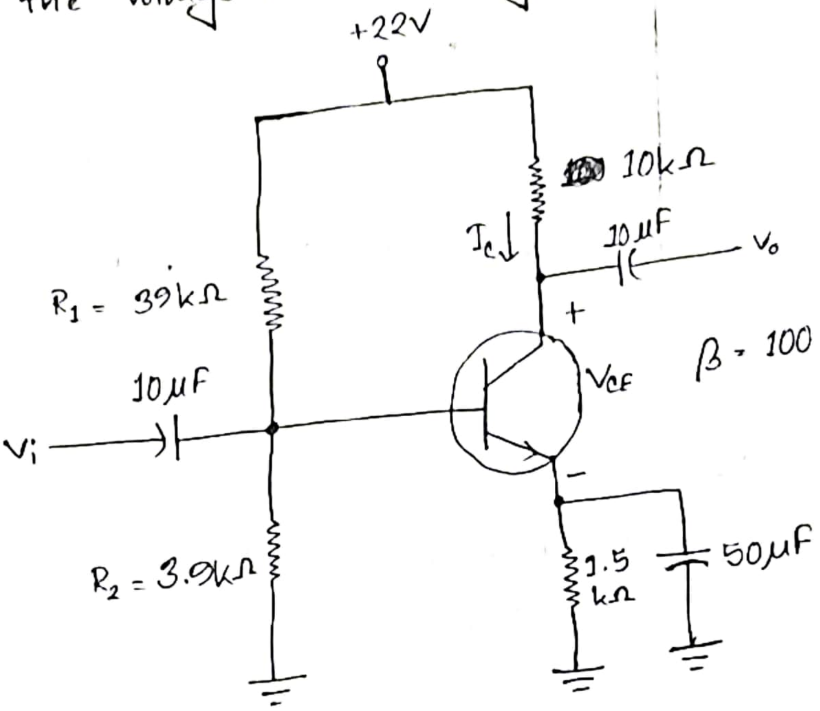
$I_B = 15 \times 10^{-3} A$

$$\therefore R_B = \frac{18 - 0.7 - (220 + 1) \times 1.1 \times 15 \times 10^{-3}}{15 \times 10^{-3}}$$

$$= 910.23 \, k\Omega$$

Example:
4.8

Determine the dc bias voltage V_{CE} and the current I_C for the voltage-divider configuration of figure.



Solution:-

$$V_{th} = \frac{R_2}{R_1 + R_2} \times V_{cc}$$

$$= \frac{3.9}{39 + 3.9} \times 22$$

$$= 2V$$

$$R_{th} = R_1 \| R_2 = \frac{R_1 \times R_2}{R_1 + R_2}$$

$$= \frac{39 \times 3.9}{39 + 3.9} = 3.54k\Omega$$

Applying KVL at input side,

$$-V_{IL} + I_B R_{IL} + V_{BE} + I_E R_E = 0$$

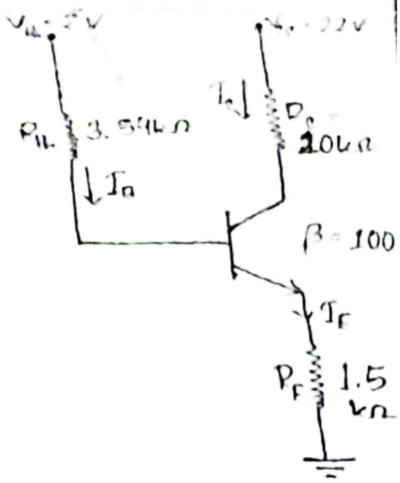
$$\Rightarrow -V_{IL} + I_B R_{IL} + V_{BE} + I_B(\beta+1)R_E = 0$$

$$\Rightarrow I_B = \frac{V_{IL} - V_{BE}}{R_{IL} + (\beta+1)R_E}$$

$$\Rightarrow I_B = \frac{2 - 0.7}{3.54 + (100+1)1.5}$$

$$\therefore I_B = 8.38\ \mu A$$

$$I_C = \beta I_B = 100 \times 8.38 = 838\ \mu A$$
$$= 838 \times 10^{-3}\ mA$$



Applying KVL at output side,

$$-V_{CC} + I_C R_C + V_{CE} + I_E R_E = 0$$

$$\Rightarrow -V_{CC} + I_C R_C + V_{CE} + I_C R_E = 0$$

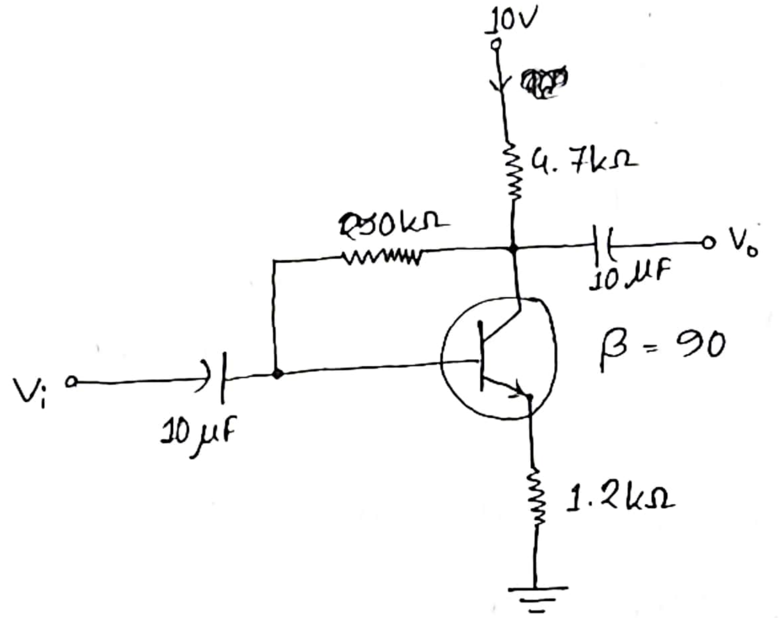
$$\Rightarrow V_{CE} = V_{CC} - I_C(R_C + R_E)$$

$$\Rightarrow V_{CE} = 22 - 838 \times 10^{-3}(10 + 1.5)$$

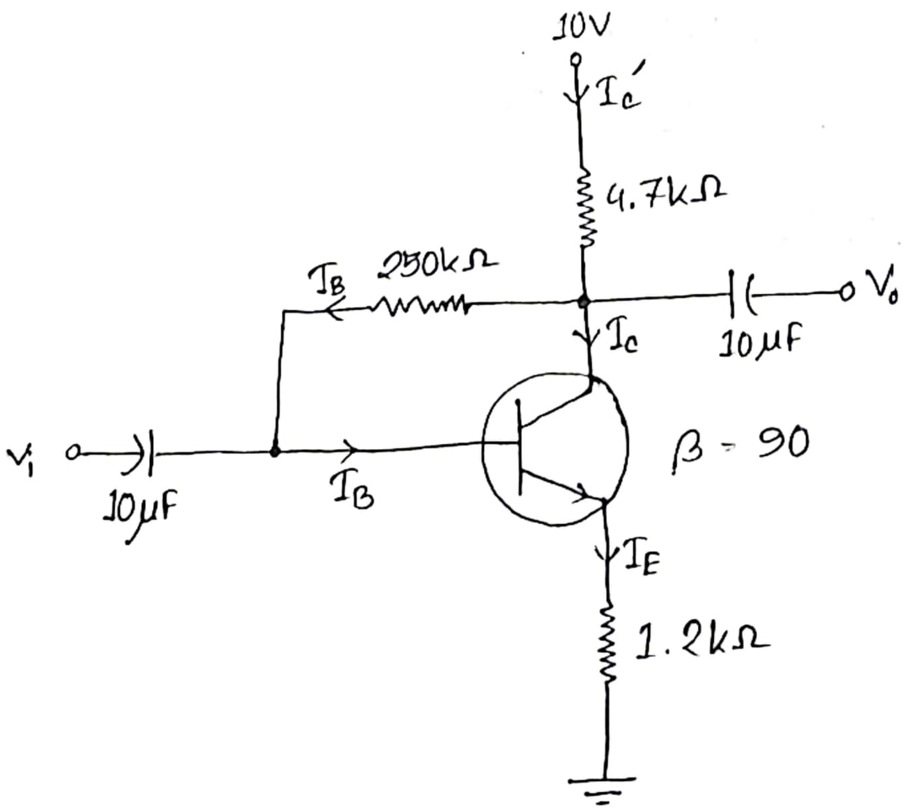
$$\therefore V_{CE} = 12.36\ V$$

Example: 4.12

Determine the quiescent levels of I_{CQ} and V_{CEQ} for the network of figure.



Solution:-



$$I_c' = I_c + I_B$$

$$I_c' = I_c = \beta I_B$$

$$I_E = I_c$$

Applying KVL at input side,

$$-V_{cc} + I_c' R_c + I_B R_B + V_{BE} + I_E R_E = 0$$

$$\Rightarrow -V_{cc} + (I_c + I_B) R_c + I_B R_B + V_{BE} + (\beta+1) I_B R_E = 0$$

$$\Rightarrow -V_{cc} + I_c R_c + I_B R_c + I_B R_B + V_{BE} + (\beta+1) I_B R_E = 0$$

$$\Rightarrow -V_{cc} + \beta I_B R_c + I_B R_c + I_B R_B + V_{BE} + (\beta+1) I_B R_E = 0$$

$$\Rightarrow -V_{cc} + I_B \left(\beta R_c + R_c + R_B + (\beta+1) R_E \right) + V_{BE} = 0$$

$$\therefore I_B = \frac{V_{cc} - V_{BE}}{\beta R_c + R_c + R_B + (\beta+1) R_E}$$

$$= \frac{10 - 0.7}{R_c(\beta+1) + R_B + (\beta+1) R_E}$$

$$= \frac{10 - 0.7}{(\beta+1)(R_c + R_E) + R_B}$$

$$= \frac{10 - 0.7}{(90+1)(4.7+1.2) + 250}$$

$$= 0.0118\,mA$$

$$\therefore I_c = 90 \times 0.0118 = 1.062\,mA$$

Applying KVL at output side,

$$-V_{cc} + I_c' R_c + V_{CE} + I_E R_E = 0$$

$$\Rightarrow -V_{cc} + I_c R_c + V_{CE} + I_c R_E = 0$$

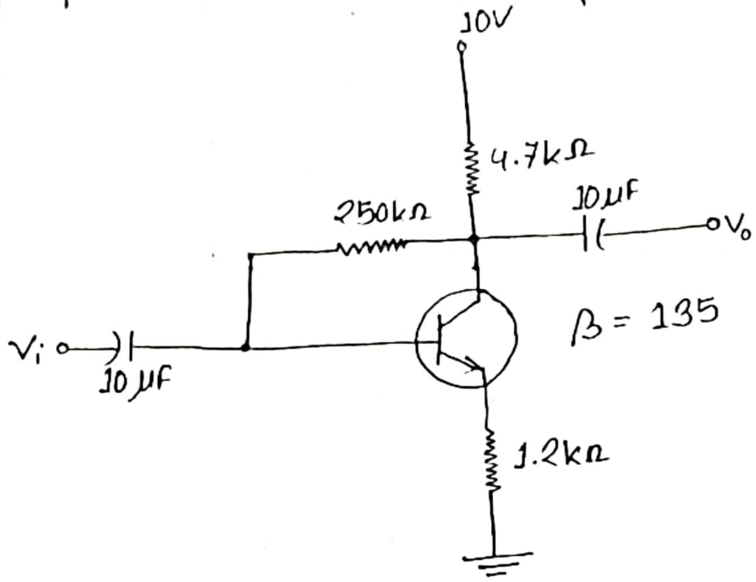
$$\Rightarrow V_{CE} = 10 - I_c (R_c + R_E)$$

$$\therefore V_{CE} = 10 - 1.062(4.7 + 1.2)$$

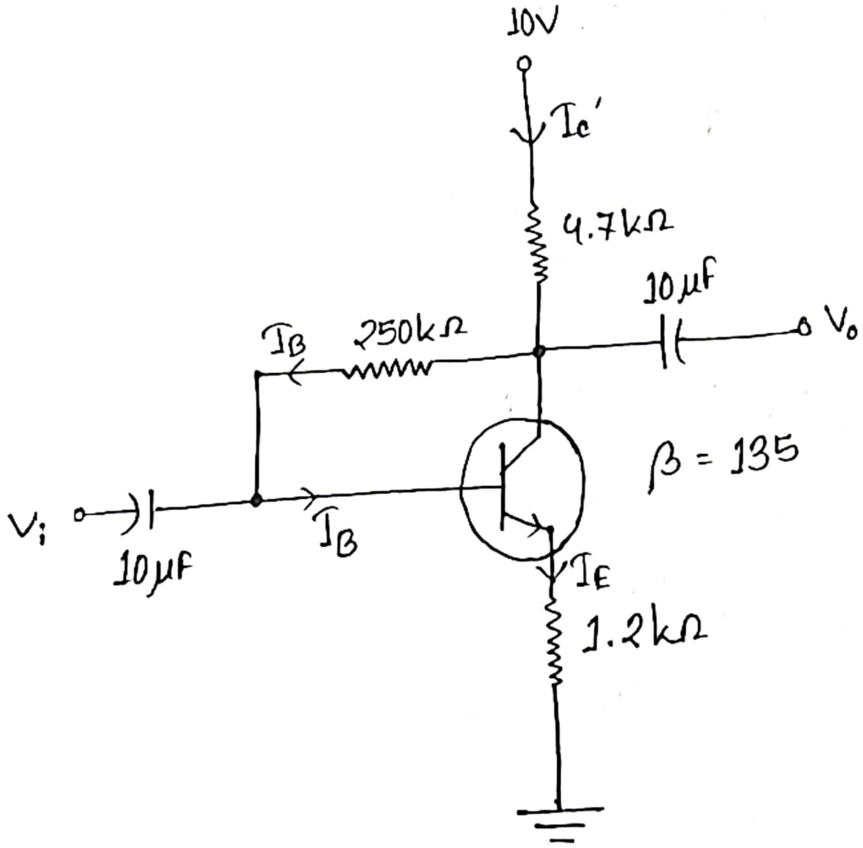
$$= 3.73\ V$$

Example 4.13

Repeat Example 4.12 using a beta of 135.



Solution:-



Applying KVL at input side,

$$-V_{CC} + I_C' R_C + I_B R_B + V_{BE} + I_E R_E = 0$$

$$\Rightarrow -V_{CC} + (I_C + I_B) R_C + I_B R_B + V_{BE} + (\beta+1) I_B R_E = 0$$

$$\Rightarrow -V_{CC} + I_C R_C + I_B R_C + I_B R_B + V_{BE} + (\beta+1) I_B R_E = 0$$

$$\Rightarrow -V_{CC} + \beta I_B R_C + I_B R_C + I_B R_B + V_{BE} + (\beta+1) I_B R_E = 0$$

$$\Rightarrow -V_{CC} + I_B \left(\beta R_C + R_C + R_B + (\beta+1) R_E\right) + V_{BE} = 0$$

$$\therefore I_B = \frac{V_{CC} - V_{BE}}{\beta R_C + R_C + R_B + (\beta+1) R_E}$$

$$= \frac{V_{CC} - V_{BE}}{(\beta+1) R_C + R_B + (\beta+1) R_E}$$

$$= \frac{V_{CC} - V_{BE}}{(\beta+1)(R_C + R_E) + R_B}$$

$$= \frac{10 - 0.7}{(135+1)(4.7+1.2) + 250}$$

$$= 8.836\ \mu A$$

$$\therefore I_C = \beta I_B = 135 \times 8.836 = 1192.86\ \mu A$$
$$= 1.19\ mA$$

Applying KVL at output side,

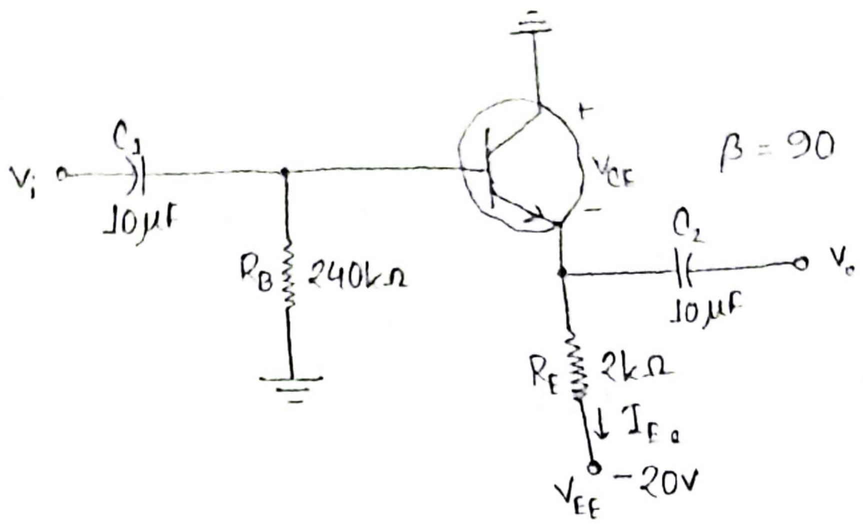
$$-V_{CC} + I_C' R_C + V_{CE} + I_E R_E = 0$$

$$\Rightarrow -V_{CC} + I_C R_C + V_{CE} + I_C R_E = 0$$

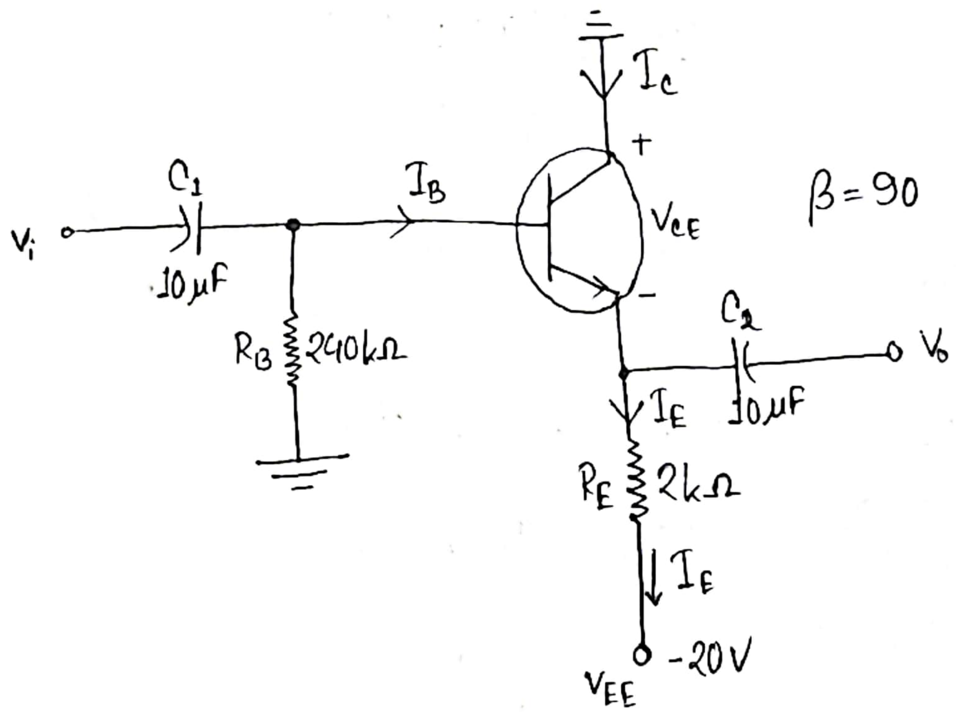
$$\therefore V_{CE} = V_{CC} - I_C (R_C + R_E)$$
$$= 10 - 1.19 (4.7 + 1.2)$$
$$= 2.979 V$$

Example 4.16

Determine V_{CE_Q} and I_{E_Q} for the network of figure.



Solution:-



Applying KVL at input side,

$$I_B R_B + V_{BE} + I_E R_E - V_{EE} = 0$$

$$\Rightarrow I_B R_B + V_{BE} + (\beta+1) I_B R_E - V_{EE} = 0$$

$$\Rightarrow I_B \left(R_B + (\beta+1) R_E\right) = V_{EE} - V_{BE}$$

$$\therefore I_B = \frac{V_{EE} - V_{BE}}{R_B + (\beta+1) R_E}$$

$$= \frac{20 - 0.7}{240 + (90+1) \times 2}$$

$$= 0.0457 \text{ mA}$$

$$I_E = (\beta+1) I_B$$

$$= (90+1) \times 0.0457$$

$$= 4.16 \text{ mA}$$

Applying KVL at output side,

$$V_{CE} + I_E R_E - V_{EE} = 0$$

$$\Rightarrow V_{CE} = V_{EE} - I_E R_E$$

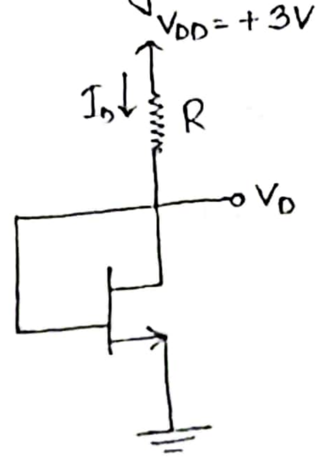
$$\therefore V_{CE} = 20 - (4.16 \times 2)$$

$$= 11.68 \text{ V}$$

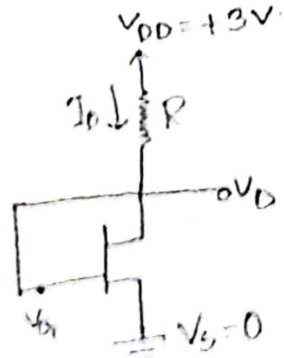
MOSFET

Example : 01

In the following NMOS, $I_D = 0.16\,mA$, $V_t = 0.6\,V$, $K_n' = 200\,\mu A/V^2$, $L = 0.4\,\mu m$ and $W = 4\,\mu m$. Find the value of R neglecting channel length and modulation effect.



Solution:-



$I_D = 0.16\,mA$
$= 0.16 \times 10^{-3}$ A
$V_t = 0.6\,V$
$K_n' = 200\,\mu A/V^2$
$L = 0.4\,\mu m$
$W = 4\,\mu m$

$V_{GS} = V_G - V_S = V_G$

$V_{DS} = V_D - V_S = V_D$

$\therefore V_G = V_D$

$\therefore V_{DS} \geq V_{GS} - V_T$ [Saturation mode]

$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_T)^2$$

$$\Rightarrow (V_{GS} - V_T)^2 = \frac{2I_D}{\mu_n C_{ox}\frac{W}{L}}$$

$$\Rightarrow (V_{GS} - V_T)^2 = \frac{2 \times 0.16 \times 10^{-3}}{200 \times 10^{-6} \times \frac{4}{0.4}} = 0.16$$

$$V_{GS} - V_T = \pm 0.4$$

$V_{GS} = 0.4 + V_t$
$= 0.4 + 0.6$
$= 1\ V$

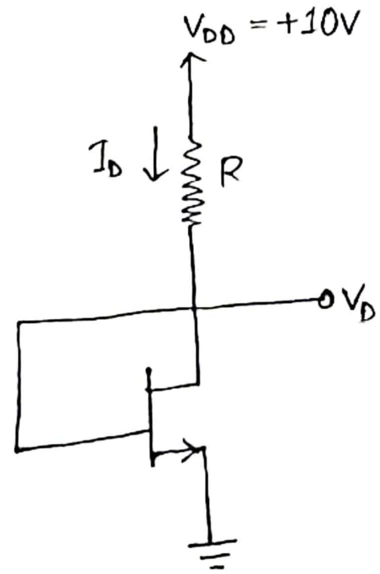
$\because V_{DS} = V_D - V_S$
$\therefore V_{DS} = V_D = 1V$

$V_{GS} = V_{DS} = 1V$

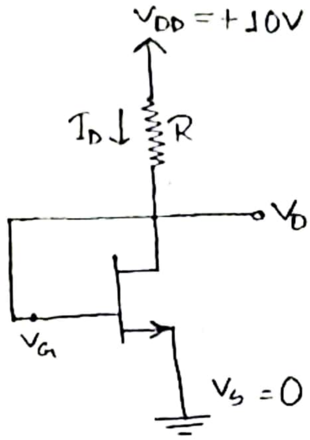
$\therefore R = \dfrac{V_{DD} - V_D}{I_D} = \dfrac{3-1}{0.16 \times 10^{-3}} = 12500\,\Omega = 12.5\,k\Omega$

Example:
02

In the following NMOS, $I_D = 0.4mA$, $V_t = 2V$, $\mu_n C_{ox} = 20\ \mu A/V^2$, $L = 10\ \mu m$ and $W = 100\mu m$. Find the value of R neglecting channel length and modulation effect.



Solution:-



Here,

$I_D = 0.4mA = 0.4 \times 10^{-3} A$

$V_t = 2V$

$\mu_n C_{ox} = 20 \mu A/V^2$

$L = 10 \mu m$

$W = 100 \mu m$

$V_{GS} = V_G - V_S = V_G$

$V_{DS} = V_D - V_S = V_D$

$\therefore V_D = V_G$

$\therefore V_{DS} \geq V_{GS} - V_T$ [Saturation Mode]

$\therefore I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_T)^2$

$\Rightarrow (V_{GS} - V_T)^2 = \frac{2 I_D}{\mu_n C_{ox} \frac{W}{L}}$

$\Rightarrow (V_{GS} - V_T)^2 = \frac{2 \times 0.4 \times 10^{-3}}{20 \times 10^{-6} \times \frac{100}{10}} = 4$

$\therefore V_{GS} - V_T = \pm 2$

$\therefore V_{GS} = 2 + V_T = 2 + 2 = 4V$

$\because V_{DS} = V_D - V_S$

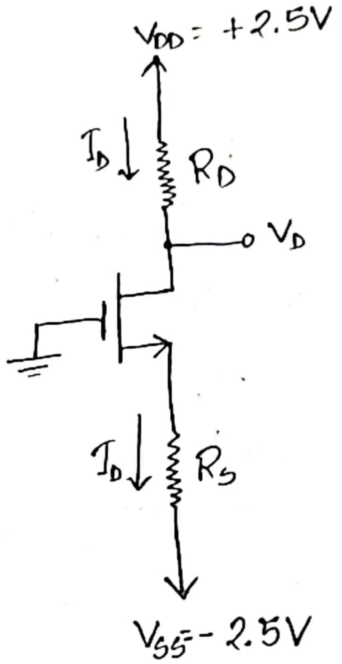
$\Rightarrow V_{DS} = V_D = 4V$

$\therefore V_{DS} = V_{GS} = 4V$

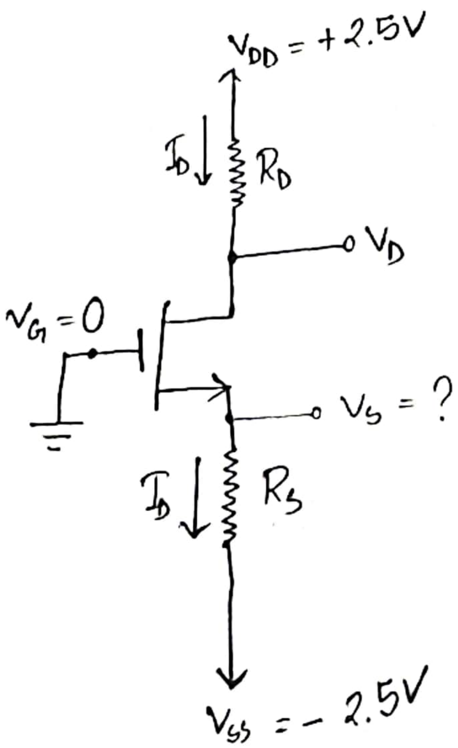
$\therefore R = \frac{V_{DD} - V_D}{I_D} = \frac{10 - 4}{0.4 \times 10^{-3}} = 15000 \Omega = 15 k\Omega$

Example: 03

Design the circuit of figure, that is, determine the values of R_D and R_S, so that the transistor operates at $I_D = 0.4mA$ and $V_D = +0.5V$. The NMOS transistor has $V_t = 0.7V$, $\mu_n C_{ox} = 100 \mu A/V^2$, $L = 1 \mu m$ and $W = 32 \mu m$. Neglect the channel-length modulation effect.



Solution:-



$I_D = 0.4mA$
$= 0.4 \times 10^{-3} A$

$V_D = +0.5V$

$V_t = 0.7V$

$\mu_n C_{ox} = 100 \mu A/V^2$

$L = 1 \mu m$

$W = 32 \mu m$

$$R_D = \frac{V_{DD} - V_D}{I_D}$$

$$= \frac{2.5 - 0.5}{0.4} = 5k\Omega$$

V_{DS}

$\rightarrow V_D - \cancel{V_S}$

$= 0.5$

$V_{GS} - V_T$

$\rightarrow V_G - \cancel{V_S} - V_T$

$= 0 - 0.7$

$= -0.7$

$V_{DS} > V_{GS} - V_T$ [Saturation Mode]

$$\therefore I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_T)^2$$

$$\Rightarrow (V_{GS} - V_T)^2 = \frac{2 I_D}{\mu_n C_{ox} \frac{W}{L}}$$

$$\Rightarrow (V_{GS} - V_T)^2 = \frac{2 \times 0.4 \times 10^{-3}}{100 \times 10^{-6} \times \frac{32}{1}} = 0.25$$

$$\therefore V_{GS} - V_T = \pm 0.5$$

$$\therefore V_{GS} = 0.5 + V_T = 0.5 + 0.7 = 1.2V$$

$$V_{GS} = V_G - V_S$$

$$\Rightarrow V_S = V_G - V_{GS} = 0 - 1.2 = -1.2V$$

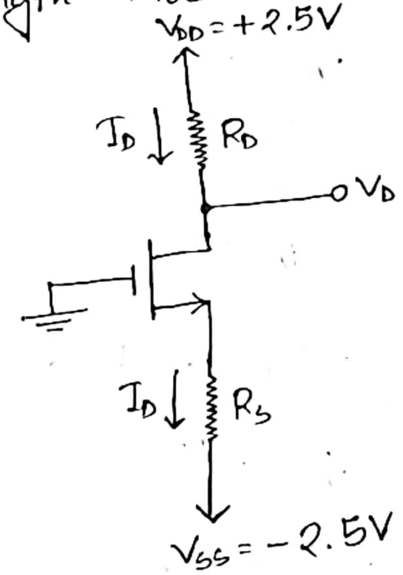
$$\therefore V_S - I_D R_S = V_{SS}$$

$$\Rightarrow R_S = \frac{V_S - V_{SS}}{I_D}$$

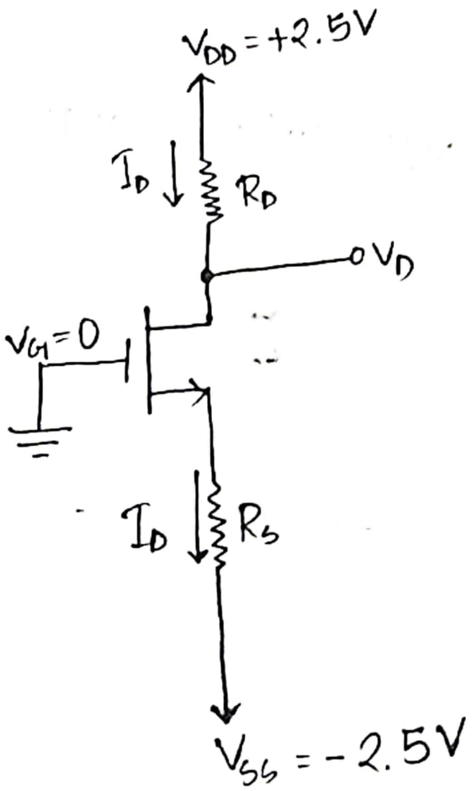
$$\therefore R_S = \frac{-1.2 + 2.5}{0.4} = 3.25k\Omega$$

Example: 04

A NMOS transistor in the circuit of figure, has $R_D = 5k\Omega$ and $R_S = 3.25k\Omega$. The transistor operates at $I_D = 0.4mA$ and $V_D = +0.5V$. It has $V_t = 0.7V$, $\mu_n C_{ox} = 100\ \mu A/V^2$, $L = 1\ \mu m$. Determine the value of gate width, W. Neglect the channel-length modulation effect.



Solution:-



Here,

$I_D = 0.4mA = 0.4 \times 10^{-3} A$
$V_D = +0.5V$
$V_t = 0.7V$
$\mu_n C_{ox} = 100 \mu A/V^2$
$L = 1 \mu m$

$R_D = 5k\Omega$
$R_S = 3.25k\Omega$

V_{DS}
$\Rightarrow V_D - V_S$
$\Rightarrow 0.5$

$V_{GS} - V_T$
$\Rightarrow V_G - V_S - V_T$
$\Rightarrow 0 - 0.7$
$= 0 - 0.7$

$\therefore V_{DS} > V_{GS} - V_T$ [Saturation Mode]

$\therefore I_D = \cancel{\frac{1}{2}} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_T)^2$
$\Rightarrow \cancel{\mu_n C_{ox} \frac{W}{L} (V_{GS} - V_T)^2 = 2I_D}$

$V_S - I_D R_S = V_{SS}$
$\Rightarrow V_S = V_{SS} + I_D R_S$
$\Rightarrow V_S = -2.5 + (0.4 \times 10^{-3}) \times (3.25 \times 10^3)$

$\therefore V_S = -1.2V$

$V_{GS} = V_G - V_S$
$= 0 - (-1.2) = 1.2V$

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Now,

$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS}-V_T)^2$$

$$\Rightarrow \mu_n C_{ox}\frac{W}{L}(V_{GS}-V_T)^2 = 2I_D$$

$$\Rightarrow \frac{W}{L}(V_{GS}-V_T)^2 = \frac{2I_D}{\mu_n C_{ox}}$$

$$\Rightarrow W = \frac{2I_D L}{\mu_n C_{ox}(V_{GS}-V_T)^2}$$

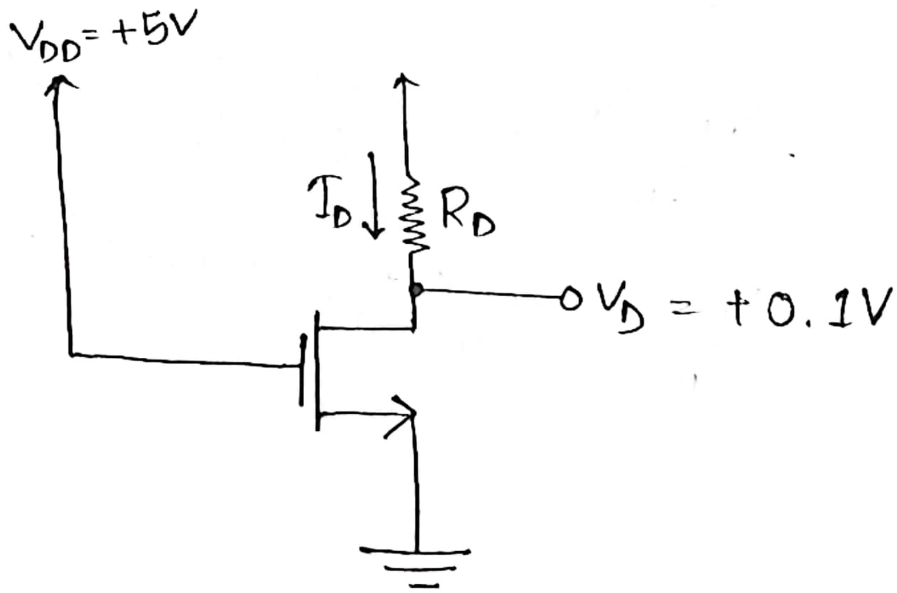
$$\Rightarrow W = \frac{2\times 0.4\times 10^{-3}\times 1\times 10^{-6}}{100\times 10^{-6}\times(1.2-0.7)^2}$$

$$\therefore W = 3.2\times 10^{-5}\,m = 32\,\mu m$$

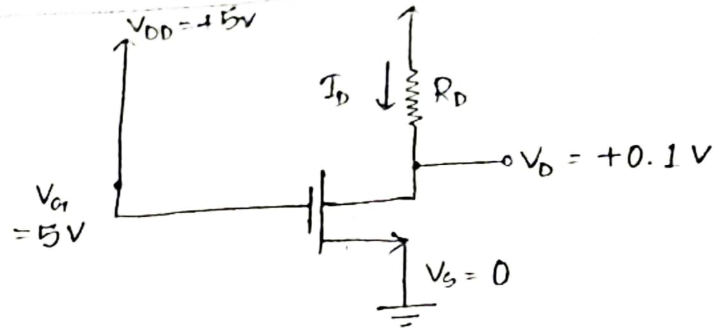
(Ans:)

Example: 05

Design the circuit in figure, to establish a drain voltage of 0.1V. What is the effective resistance between drain and source at this operating point? Let and $V_t = 1V$ and $k_n'(W/L) = 1\,mA/V^2$.



Solution:-



$V_t = 1\,V; \quad k_n'(W/L) = 1\,mA/V^2$

V_{DS}

$\Rightarrow V_D - \cancel{V_S}$

~~0.1 - 0~~

$= 0.1$

$V_{GS} - V_T$

$= V_G - \cancel{V_S} - V_T$

$= 5 - 1$

$= 4$

$V_{DS} < V_{GS} - V_T$ [Triode Mode]

$$\therefore I_D = \mu_n C_{ox} \frac{W}{L}\left[(V_{GS} - V_T)V_{DS} - \frac{1}{2}V_{DS}^2\right]$$

$$= 1 \times 10^{-3}\left[(5-1)\times 0.1 - \frac{1}{2}(0.1)^2\right]$$

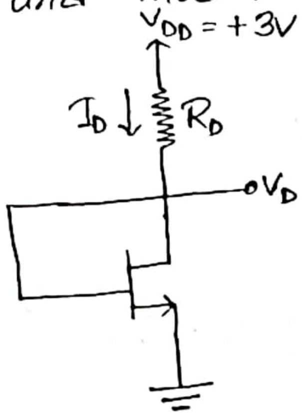
$$= 3.95 \times 10^{-4} = 0.395\,mA$$

$$R_D = \frac{V_{DD} - V_D}{I_D} = \frac{5 - 0.1}{0.395} = 12.4\,k\Omega$$

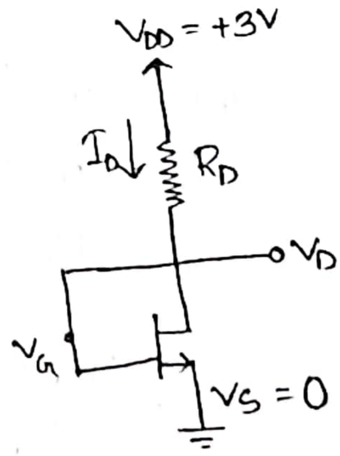
$$R_{DS} = \frac{V_{DS}}{I_D} = \frac{0.1}{0.395} = 0.25\,k\Omega$$

Example : 06

Design the circuit in figure to obtain a current I_D of $80\,\mu A$. Find the value required for R and find the dc voltage V_D. Let the NMOS transistor have $V_t = 0.6V$, $\mu_n C_{ox} = 200\,\mu A/V^2$, $L = 0.8\,\mu m$ and $W = 4\,\mu m$. Neglect the channel length and modulation effect.



Solution:-



Here,

$I_D = 80\,\mu A = 80 \times 10^{-6} A$

$V_t = 0.6V$

$\mu_n C_{ox} = 200\,\mu A/V^2$

$L = 0.8\,\mu m$

$W = 4\,\mu m$

$V_{DS} = V_D - V_S = V_D$

$V_{GS} = V_G - V_S = V_G$

$V_D = V_G$

$\therefore V_{DS} \geq V_{GS} - V_T$ [Saturation Mode]

$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS}-V_T)^2$$

$$\Rightarrow (V_{GS}-V_T)^2 = \frac{2I_D}{\mu_n C_{ox}\frac{W}{L}}$$

$$\Rightarrow (V_{GS}-V_T)^2 = \frac{2\times 80\times 10^{-6}}{200\times 10^{-6}\times \frac{4}{0.8}}$$

$$\Rightarrow (V_{GS}-V_T)^2 = 0.16$$

$$\therefore V_{GS}-V_T = \pm 0.4$$

$$\therefore V_{GS} = 0.4 + V_T = 0.4 + 0.6 = 1V$$

$$\therefore V_{GS} = V_{DS} = 1V$$

$$V_{DS} = V_D - V_S$$

$$\Rightarrow V_D = V_{DS} + V_S = 1 + 0 = 1V$$

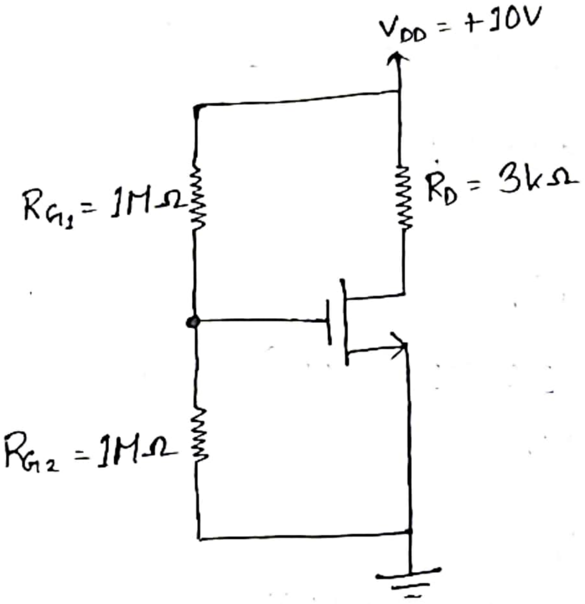
$$\therefore V_D = 1V$$

$$R = \frac{V_{DD} - V_D}{I_D}$$

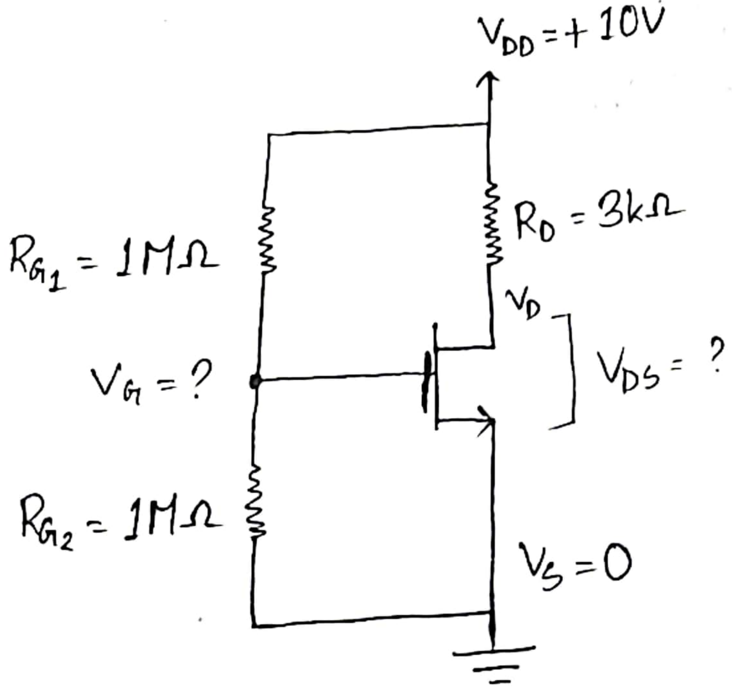
$$= \frac{3-1}{80\times 10^{-6}} = 25000\,\Omega = 25k\Omega$$

Example : 07

Find I_{Dmax} that can flow through the ideal N-MOSFET. Also find V_{DS} at I_{Dmax}. Given $V_{tn} = 1.5V$ and $K_n'(W/L) = 2.76 \times 10^{-6} A/V^2$. Find the power dissipated in the MOSFET.



Solution:-



$$V_G = \frac{R_{G_2}}{R_{G_1} + R_{G_2}} \times V_{DD}$$

$$= \frac{1}{1+1} \times 10 = 5V$$

$$V_{GS} = V_G - V_S$$
$$= 5 - 0 = 5V$$

Lets assume the MOSFET in Saturation Mode,

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{tn})^2$$

$$= \frac{1}{2} \times 2.76 \times 10^{-6} \times (5 - 1.5)^2$$

$$= 1.69 \times 10^{-5}\ A$$

$$\therefore V_D = V_{DD} - I_D R_D$$

$$= 10 - (1.69 \times 10^{-5} \times 3)$$

$$= 9.99\ V$$

$$\therefore V_{DS} = V_D - V_S$$

$$= 9.99 - 0 = 9.99V$$

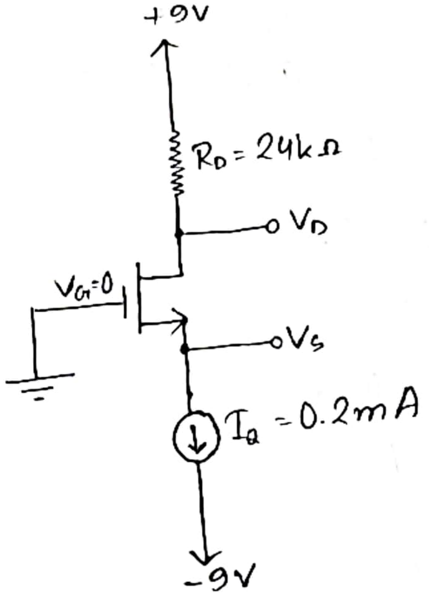
$$P_D = V_{DS} I_D$$

$$= 9.99 \times 1.69 \times 10^{-5}$$

$$= 1.68 \times 10^{-4}\ W$$

Example : 08

In the following NMOS Transistor $V_t = 0.6\,V$ and $K_n'(W/L) = 200\ \mu A/V^2$. Determine the value of V_S and V_D.



Solution:-

$$V_D = 9 - I_D R_D$$
$$= 9 - (0.2 \times 10^{-3} \times 24 \times 10^3)$$
$$= 4.2V$$

V_{DS}	$V_{GS} - V_T$
$V_D - \cancel{V_S}$	$V_G - \cancel{V_S} - V_T$
$= 4.2$	$0 - 0.6$
	$= -0.6$

$\therefore V_{DS} > V_{GS} - V_T$ [Saturation Mode]

$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS}-V_T)^2$$

$$\Rightarrow (V_{GS}-V_T)^2 = \frac{2I_D}{\mu_n C_{ox}\frac{W}{L}}$$

$$\Rightarrow (V_{GS}-V_T)^2 = \frac{2\times 0.2\times 10^{-3}}{200\times 10^{-6}}$$

$$\Rightarrow (V_{GS}-V_T)^2 = 2$$

$$\therefore V_{GS}-V_T = \pm 1.41$$

$$\therefore V_{GS} = 1.41 + V_T$$

$$= 1.41 + 0.6$$

$$= 2.01\text{ V}$$

$$\therefore V_{GS} = V_G - V_S$$

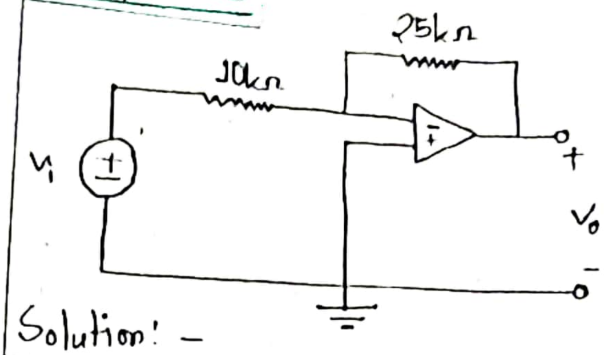
$$\Rightarrow V_S = V_G - V_{GS}$$

$$\therefore V_S = 0 - 2.01$$

$$= -2.01\text{V}$$

OP-AMP

Example: 5.3



Refere to the op amp in figure. If $v_i = 0.5V$, calculate (a) the output voltage v_o, and (b) the current in the 10 kΩ resistor.

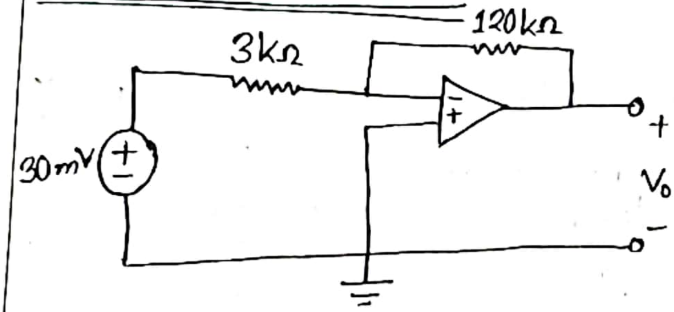
Solution:-

(a) $V_o = -\frac{R_f}{R_1} \times V_i$

$= -\frac{25}{10} \times 0.5 = -1.25\ V$

(b) $i = \frac{V_i - 0}{R_1} = \frac{0.5 - 0}{10 \times 10^3} = 50\ \mu A$

Practice Problem: 5.3



Find the output of the op amp circuit shown in figure. Calculate the current through the feedback resistor.

Solution:-

$V_o = -\frac{R_f}{R_1} \times V_i$

$= -\frac{120}{3} \times 30$

$= -1200\ mV$

$= -1.2\ V$

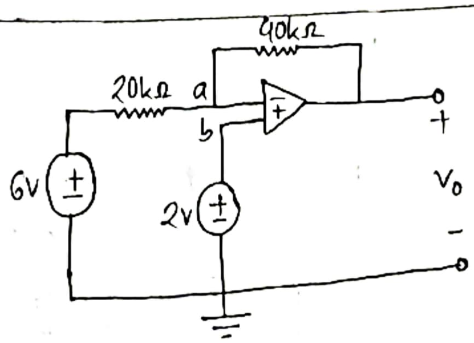
$i = \frac{0 - V_o}{R_f}$

$= \frac{0 - (-1.2)}{120 \times 10^3}$

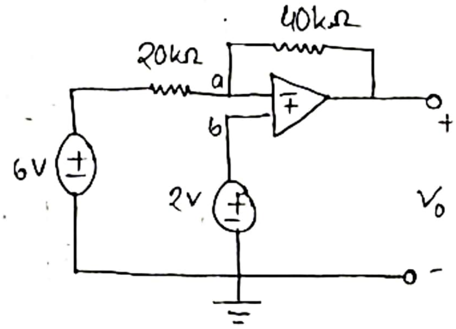
$= 10\ \mu A$

Example: 5.4

Determine V_o in the op amp circuit shown in figure.



Solution:-



Applying KCL at node a,

$$\frac{V_i - V_a}{R_i} = \frac{V_a - V_o}{R_f}$$

$$\Rightarrow \frac{6-2}{20} = \frac{2-V_o}{40}$$

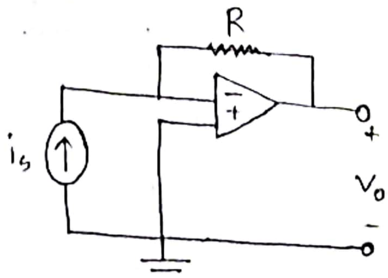
$$\Rightarrow 160 = 40 - 20V_o$$

$$\Rightarrow -20V_o = 120$$

$$\therefore V_o = -6V$$

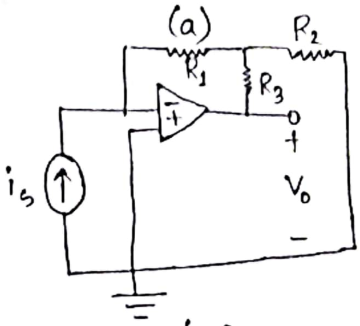
Practice Problem: 5.4

Two kinds of current-to-voltage converters (also known as transresistance amplifiers) are shown in figure.



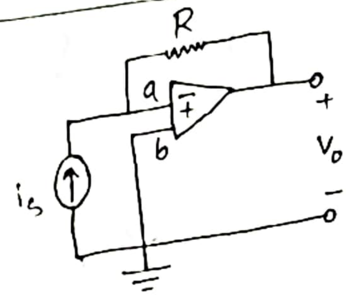
(a) Show that for the converter in figure (a),

$$\frac{V_o}{i_s} = -R$$



(b) Show that for the converter in figure (b);

$$\frac{V_o}{i_s} = -R_1\left(1 + \frac{R_3}{R_1} + \frac{R_3}{R_2}\right)$$



Solution :-

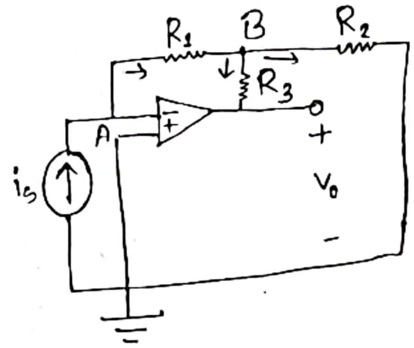
(a) Applying KCL at node a,

$$i_s = \frac{V_a - V_o}{R}$$

$$\Rightarrow i_s = \frac{0 - V_o}{R}$$

$$\Rightarrow -R = \frac{V_o}{i_s}$$

$$\therefore \frac{V_o}{i_s} = -R \quad \text{(showed)}$$



(b) Applying KCL at node A,

$$i_s = 0 + \frac{V_A - V_B}{R_1}$$

$$\Rightarrow i_s = \frac{0 - V_B}{R_1}$$

$$\Rightarrow i_s = -\frac{V_B}{R_1}$$

$$\therefore V_B = -i_s R_1$$

Applying KCL at node B,

$$\frac{V_A - V_B}{R_1} = \frac{V_B - V_o}{R_3} + \frac{V_B - 0}{R_2}$$

$$\Rightarrow \frac{i_s R_1}{R_1} = \frac{-i_s R_1 - V_o}{R_3} + \frac{-i_s R_1}{R_2}$$

$$\Rightarrow \frac{i_s \cancel{R_1}}{\cancel{R_1}} = -\frac{i_s R_1}{R_3} - \frac{V_o}{R_3} - \frac{i_s R_1}{R_2}$$

$$\Rightarrow \frac{V_o}{R_3} = -i_s - \frac{i_s R_1}{R_3} - \frac{i_s R_1}{R_2}$$

$$\Rightarrow V_o = -i_s R_3 - i_s R_1 - \frac{i_s R_1 R_3}{R_2}$$

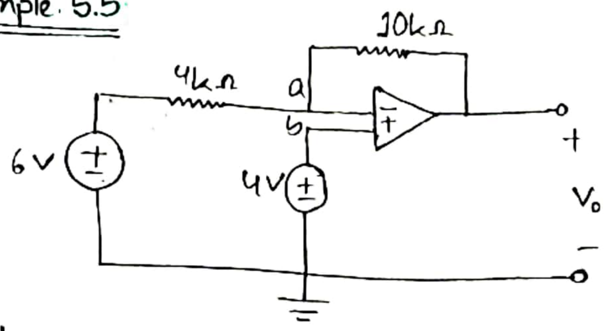
$$\Rightarrow \frac{V_o}{i_s} = -R_3 - R_1 - \frac{R_1 R_3}{R_2}$$

$$\Rightarrow \frac{V_o}{i_s} = -R_1\left(\frac{R_3}{R_1} + 1 + \frac{R_3}{R_2}\right)$$

$$\therefore \frac{V_o}{i_s} = -R_1\left(1 + \frac{R_3}{R_1} + \frac{R_3}{R_2}\right)$$

(showed)

Example: 5.5



For the op amp circuit in figure, calculate the output voltage V_o.

Solution:-

Applying KCL at node a,

$$\frac{V_i - V_a}{R_1} = \frac{V_a - V_o}{R_f}$$

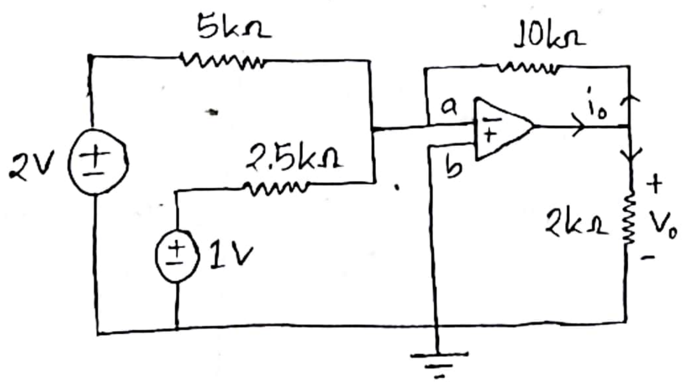
$$\Rightarrow \frac{6-4}{4} = \frac{4-V_o}{10}$$

$$\Rightarrow 20 = 16 - 4V_o$$

$$\Rightarrow -4V_o = 4$$

$$\therefore V_o = -1\,V \quad \text{(Ans:)}$$

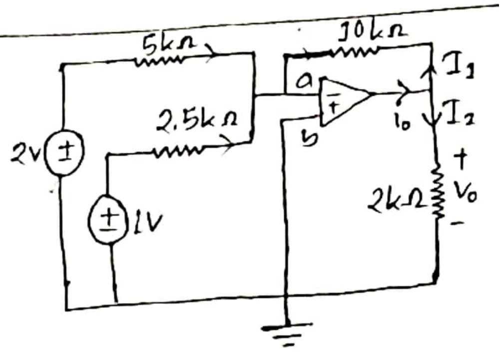
Example: 5.6



Calculate v_o and i_o in the op amp circuit in figure

Solution :-

~~Applying KCL at node a,~~



$$V_o' = -\frac{10}{5} \times 2 = -4V$$

$$V_o'' = -\frac{10}{2.5} \times 1 = -4V$$

$$\therefore V_o = V_o' + V_o'' = -4 - 4 = -8V$$

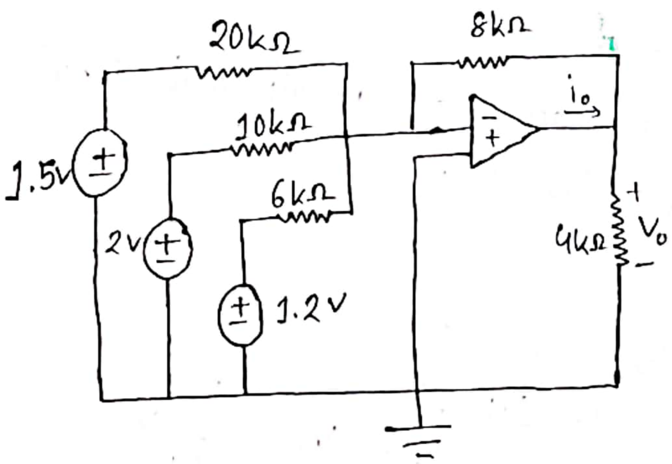
Applying KCL at output,

$$i_o = I_1 + I_2$$

$$= \frac{V_o - 0}{10} + \frac{V_o - 0}{2}$$

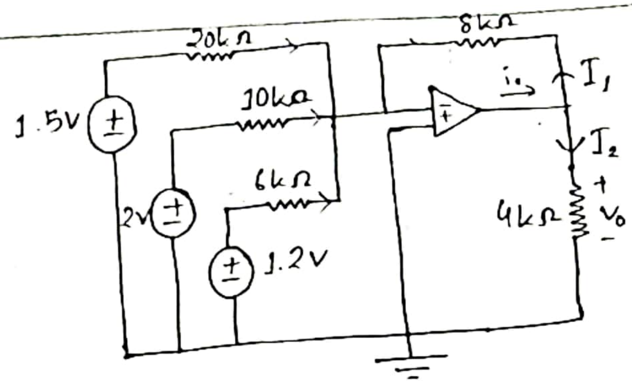
$$= \frac{-8}{10} + \frac{-8}{2} = -4.8mA$$

Practice Problem: 5.6



Find V_o and i_o in the op amp circuit shown in figure.

Solution:-



$$V_o' = -\frac{8}{20} \times 1.5$$
$$= -0.6\,V$$

$$V_o'' = -\frac{8}{10} \times 2$$
$$= -1.6\,V$$

$$V_o''' = -\frac{8}{6} \times 1.2 = -1.6\,V$$

$$\therefore V_o = -0.6 - 1.6 - 1.6 = -3.8\,V$$

Applying KCL at output,

$$i_o = I_1 + I_2$$
$$= \frac{V_o - 0}{8} + \frac{V_o - 0}{4}$$
$$= \frac{-3.8}{8} + \frac{-3.8}{4} = -1.425\,mA.$$

Example: 5.7

Design an op amp circuit with inputs v_1 and v_2 such that $V_o = -5v_1 + 3v_2$.

Solution:-

$$V_o = 3v_2 - 5v_1$$

$$\therefore V_o = \frac{R_2\left(1 + \frac{R_1}{R_2}\right)}{R_1\left(1 + \frac{R_3}{R_4}\right)} V_2 - \frac{R_2}{R_1} V_1$$

Here,

$$\frac{R_2}{R_1} = 5 \qquad \Big| \qquad \frac{R_1}{R_2} = \frac{1}{5}$$

$$\Rightarrow R_2 = 5R_1$$

Now,

$$\frac{R_2\left(1+\frac{R_1}{R_2}\right)}{R_1\left(1+\frac{R_3}{R_4}\right)} = 3$$

$$\Rightarrow \frac{5\left(1+\frac{1}{5}\right)}{\left(1+\frac{R_3}{R_4}\right)} = 3$$

$$\Rightarrow \cancel{\frac{6}{\frac{R_4+R_3}{R_4}}} \Rightarrow \frac{6}{1+\frac{R_3}{R_4}} = 3$$

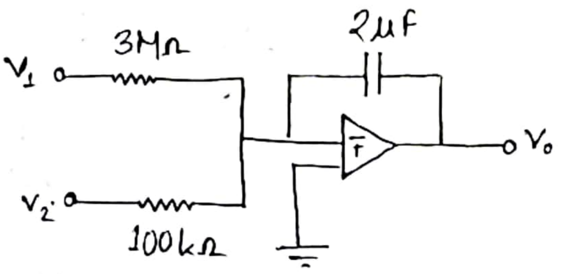
$$\Rightarrow 6 = 3\left(1+\frac{R_3}{R_4}\right)$$

$$\Rightarrow 2 = 1 + \frac{R_3}{R_4}$$

$$\therefore R_3 = R_4$$

Example: 6.13

If $v_1 = 10\cos 2t$ mV and $v_2 = 0.5t$ mV, find v_o in the op amp circuit in figure. Assume that the voltage across the capacitor is initially zero.



Solution:-

Here,

$v_1 = 10\cos 2t$ mV

$v_2 = 0.5t$ mV

$$v_o' = -\frac{1}{R_1 C}\int v_1\,dt$$

$$= -\frac{1}{3\times10^6\times2\times10^{-6}}\int_0^t 10\cos 2t\,dt$$

$$= -\frac{1}{6}\times\frac{10\sin 2t}{2}$$

$$= -0.833\sin 2t$$

$$V_o'' = -\frac{1}{R_2 C}\int v_2\,dt$$

$$= -\frac{1}{100\times 10^3 \times 2\times 10^{-6}}\int_0^t 0.5t\,dt$$

$$= -5\,.\,\frac{0.5t^2}{2}$$

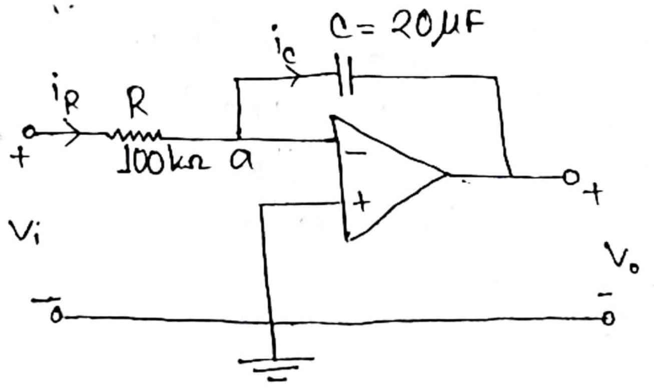
$$= -1.25\,t^2$$

$$\therefore V_o = V_o' + V_o''$$

$$= -0.833\sin 2t - 1.25t^2 \quad mV$$

Practice Problem: 6.13

The integrator in figure has $R = 100\,k\Omega$, $C = 20\,\mu F$. Determine the output voltage when a dc voltage of $10\,mV$ is applied at $t = 0$. Assume that the op amp is initially nulled.



Solution:-

$$V_o = -\frac{1}{RC}\int V_i\,dt$$

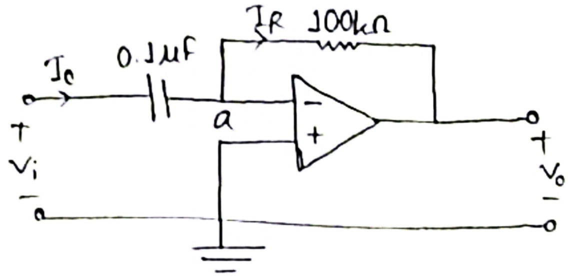
$$= -\frac{1}{100\times 10^3 \times 20\times 10^{-6}}\int_0^t 10\,dt$$

$$= -\frac{10}{2}\left[t\right]_0^t$$

$$= -5t \quad mV$$

Practice Problem: 6.14

The differentiation in figure has $R = 100k\Omega$ and $C = 0.1\mu F$. Given that $v_i = 3t$ V, determine the output v_o



Solution:-

$$V_o = -RC\frac{dv_i}{dt}$$

$$= -100\times10^3\times0.1\times10^{-6}\times\frac{d}{dt}(3t)$$

$$= -100\times10^3\times0.1\times10^{-6}\times3$$

$$= -0.03V$$

$$= -30\,mV$$

(Ans:)